

b)

- We call $v_1, v_2, \dots, v_m \in V$ linearly independent if:

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_m v_m = 0_m.$$

$$(\Leftrightarrow) \alpha_1 = \alpha_2 = \dots = \alpha_m = 0.$$

- Otherwise, the vectors are called linearly dependent.
- Linear independence $(\Leftrightarrow)$ if the det. formed from $v_1, \dots, v_m$ is not zero.
- We call $B = \langle v_1, \dots, v_m \rangle$ a basis of V if:
 - $v_1, \dots, v_m$ are linearly independent.
 - $\langle v_1, \dots, v_m \rangle = V$.

$$① \quad v_1 = (1, -1, 0)$$

$$v_2 = (2, 1, 1) \in_{\mathbb{R}} \mathbb{R}^3$$

$$v_3 = (1, 5, 2)$$

 a) v_1, v_2, v_3 are linearly dependent?

 b) v_1, v_2 linearly independent?

$$a) \quad \begin{vmatrix} 1 & 2 & 1 \\ -1 & 1 & 5 \\ 0 & 1 & 2 \\ 1 & 2 & 1 \\ -1 & 1 & 5 \end{vmatrix} = 2 - 1 + 4 - 5 = 0 \Rightarrow v_1, v_2, v_3 \text{ lin. dependent.}$$

$$b) \quad \alpha_1 v_1 + \alpha_2 v_2 = 0 \Leftrightarrow \alpha_1 (1, -1, 0) + \alpha_2 (2, 1, 1) = (0, 0, 0)$$

$$(\Leftrightarrow) \begin{cases} \alpha_1 + 2\alpha_2 = 0 \\ -\alpha_1 + \alpha_2 = 0 \\ \alpha_2 = 0 \end{cases} \quad \alpha_1 = 0 \Rightarrow v_1, v_2 \text{ are linearly indep.}$$

$$② \quad a) \quad v_1 = (1, 0, 2)$$

$$v_2 = (-1, 2, 1) \in_{\mathbb{R}} \mathbb{R}^3$$

$$v_3 = (3, 1, 1)$$

$$b) \quad v_1 = (1, 2, 3, 4)$$

$$v_2 = (2, 3, 4, 1) \in_{\mathbb{R}} \mathbb{R}^4$$

$$v_3 = (3, 4, 1, 2)$$

$$v_4 = (4, 1, 2, 3)$$

Check the linear independence of the vectors.

$$a) \begin{vmatrix} 1 & -1 & 3 \\ 0 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 3 \\ 0 & 2 & 1 \end{vmatrix} = 2 - 2 - 1 - 12 = -13 \neq 0 \Rightarrow \text{lin. independent.}$$

$$b) \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix} \xrightarrow{L_1+L_2, L_3 \rightarrow C_4} \begin{vmatrix} 1 & 2 & 3 & 10 \\ 2 & 3 & 4 & 10 \\ 3 & 4 & 1 & 10 \\ 4 & 1 & 2 & 10 \end{vmatrix} \xrightarrow{L_2, L_3, L_4 - L_1} \begin{vmatrix} 1 & 2 & 3 & 10 \\ 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 \\ 3 & 1 & 1 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & 3 & 10 \\ 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 \\ 3 & 1 & 1 & 0 \end{vmatrix} = 10 \cdot (-1)^5 \cdot \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 1 & 1 \end{vmatrix} = -10 \cdot \begin{vmatrix} 4 & 0 & 0 \\ 2 & 2 & 2 \\ 3 & 1 & 1 \end{vmatrix} =$$

$$= -10 \cdot 4 \cdot (-1)^2 \cdot \begin{vmatrix} 2 & 2 \\ -1 & -1 \end{vmatrix} = -40 (-2 + 2) = +160 \neq 0 \Rightarrow$$

$\Rightarrow v_1, v_2, v_3, v_4$ lin. indep.

$$\textcircled{3} \quad a \in \mathbb{R} \setminus \{-\sqrt{2}, 0, \sqrt{2}\}.$$

$$\textcircled{4} \quad v_1 = (1, -2, 0, -1)$$

$$v_2 = (2, 1, 1, 0) \in \mathbb{R}^4.$$

$$v_3 = (0, a, 1, 2)$$

$a \in \mathbb{R}$ s.t. v_1, v_2, v_3 linearly dep.

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0$$

$$\Leftrightarrow \alpha_1 (1, -2, 0, -1) + \alpha_2 (2, 1, 1, 0) + \alpha_3 (0, a, 1, 2) = (0, 0, 0, 0)$$

$$\Rightarrow \begin{cases} \alpha_1 + 2\alpha_2 = 0 \\ -2\alpha_1 + \alpha_2 + a \cdot \alpha_3 = 0 \\ \alpha_2 + \alpha_3 = 0 \\ -\alpha_1 + 2\alpha_3 = 0 \end{cases}$$

$$\begin{aligned} &\Rightarrow \alpha_2 = -\alpha_3 \quad \Rightarrow \alpha_1 = 2\alpha_3 \\ &\Rightarrow -4\alpha_3 - \alpha_3 + a \cdot \alpha_3 = 0 \Rightarrow -5\alpha_3 + a \cdot \alpha_3 = 0 \Rightarrow (a-5)\alpha_3 = 0. \end{aligned}$$

$$\text{if } a=5 \Rightarrow \alpha_3 = \alpha_2 = 0 = \alpha_1 \Rightarrow \text{lin. indep.}$$

$$\text{so } a \neq 5, \quad a \in \mathbb{R} \setminus \{5\}$$

$\alpha_3 \neq 0$, as v_1, v_2, v_3 - lin. indep.

$$⑤ \quad v_1 = (1, 1, 0)$$

$$v_2 = (-1, 0, 2) \in \mathbb{R}^3$$

$$v_3 = (1, 1, 1)$$

3 vectors in the canonical basis (the easiest way to write vectors that generate the whole space)

- a) Show that the list (v_1, v_2, v_3) is a basis of $\mathbb{R}^3$.
 b) Express (e_1, e_2, e_3) as a linear comb. of v_1, v_2, v_3
 c) Det. the coord. $u = (1, -1, 2)$ in both cases.

a) 1) Show linear independence

$$\begin{vmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{vmatrix} = (+1) \cdot (-1)^4 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = +1 \neq 0$$

$\Rightarrow$ lin. indep.

$$-L_1 + L_2 \rightarrow L_2$$

2) $(\forall) x \in \mathbb{R}^3 = (a, b, c), \quad x = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$

$$② \quad (a, b, c) = \alpha_1 (1, 1, 0) + \alpha_2 (-1, 0, 2) + \alpha_3 (1, 1, 1)$$

$$\begin{cases} \alpha_1 - \alpha_2 + \alpha_3 = a \\ \alpha_1 + \alpha_3 = b \Rightarrow \alpha_1 = b - \alpha_3 \Rightarrow \dots \\ 2\alpha_2 + \alpha_3 = c \Rightarrow 2\alpha_2 = c - \alpha_3 \\ \Rightarrow \alpha_2 = \frac{c - \alpha_3}{2} \end{cases}$$

$$\Rightarrow \alpha_1 = a - \alpha_3 + \alpha_2 = a - \alpha_3 + \frac{c - \alpha_3}{2}$$

$$b - \alpha_3 - \frac{c - \alpha_3}{2} + \alpha_3 = a \quad | \cdot 2$$

$$\Rightarrow 2b - 2\alpha_3 - c + \alpha_3 + 2\alpha_3 = 2a$$

$$\Rightarrow 2b - c + \alpha_3 = 2a \Rightarrow \alpha_3 = 2a - 2b + c$$

$$\alpha_1 = b - \alpha_3 = b - 2a + 2b - c \Rightarrow \alpha_1 = 3b - 2a - c$$

$$\alpha_2 = \frac{c - 2a + 2b - c}{2} = b - a \Rightarrow \alpha_2 = b - a$$

1), 2) $\Rightarrow (v_1, v_2, v_3)$ basis.

$$\alpha_1 = -b - 2a + c$$

$$b) \quad e_1 = \begin{pmatrix} a & b & c \\ 1 & 0 & 0 \end{pmatrix}$$

$$e_2 = (0, 1, 0)$$

$$e_3 = (0, 0, 1)$$

$$ex: \quad \mathbb{R}^2 = \langle e_1, e_2 \rangle = \langle (1, 0), (0, 1) \rangle$$

$$\mathbb{R} = \langle 1 \rangle$$

$$\mathbb{C} = \langle 1 \rangle$$

(all scalars can be also complex nr.)

$$\mathbb{C} = \langle 1, i \rangle$$

$\mathbb{R}$

$$\mathcal{M}_2 = \left\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\rangle$$

$$\mathbb{R}_2[x] = \langle x^2, x, 1 \rangle$$

$$\mathbb{Z}_m = \langle 1 \rangle$$

CANONICAL
BASES.

$$e_1 = \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3$$

$$a) e_1 = -2 \cdot v_1 + (-1) \cdot v_2 + 2 \cdot v_3$$

$$e_2 = 3 \cdot v_1 + 1 \cdot v_2 + \overset{(-2)}{v_3}$$

$$e_3 = (-1) \cdot v_1 + 0 \cdot v_2 + 1 \cdot v_3$$

$$(-2, -1, 1)$$

$$e_1 = (1, 0, 0) \quad \downarrow$$

$$x_1 = 3b - 2a - c = \underline{\underline{-3 + 4 - 1 = 0}}$$

$$c) \quad u = (1, -1, 2)$$

because using canonical e_1, e_2, e_3 is the easiest.

$$u_1 = (1 \cdot e_1 + (-1) \cdot e_2 + 2 \cdot e_3)$$

$$u = 1 \cdot v_1 + (-2) \cdot v_2 + 6 \cdot v_3$$

$$⑧ \quad \mathbb{R}_2[x] = \{ f \in \mathbb{R}[x] \mid \text{degree}(f) \leq 2 \}$$

$$E = (1, x, x^2) \quad \text{canonical basis (prove in the exam, maybe)}$$

$$B = (1, x-a, (x-a)^2), \quad a \in \mathbb{R}.$$

B basis for $\mathbb{R}_2[x]$?

Determine the coordinates of $f = a_0 + a_1 x + a_2 x^2$ in basis B.

$$\alpha_1 \cdot v_1 + \alpha_2 \cdot v_2 + \alpha_3 \cdot v_3 = 0$$

$$\Rightarrow \alpha_1 \cdot 1 + \alpha_2 \cdot (x-a) + \alpha_3 (x-a)^2 = 0.$$

$$\Rightarrow \alpha_1 + \alpha_2 \cdot x - \alpha_2 \cdot a + \alpha_3 \cdot x^2 - 2\alpha_3 \cdot a \cdot x + \alpha_3 \cdot a^2 = 0$$

$$\Rightarrow (\alpha_1 - a \cdot \alpha_2 + a^2 \cdot \alpha_3) + (\alpha_2 \cdot x - 2\alpha_3 \cdot a \cdot x) + \alpha_3 \cdot x^2 = 0$$

$$\Rightarrow \begin{cases} \alpha_1 = 0 \\ \alpha_2 = 0 \Rightarrow \text{lin. indep.} \\ \alpha_3 = 0 \end{cases}$$

$$(4) f \in \mathbb{R}_2[x], f = a_0 + a_1 \cdot x + a_2 \cdot x^2, \exists! \alpha_1, \alpha_2, \alpha_3 \text{ s.t.}$$

$$(\alpha_1 - a \cdot \alpha_2 + a^2 \cdot \alpha_3) + (\alpha_2 - 2a \cdot \alpha_3)x + \alpha_3 x^2 = f$$

$$\Rightarrow \begin{cases} a_0 = \alpha_1 - a \cdot \alpha_2 + a^2 \cdot \alpha_3 \Rightarrow \alpha_1 = a_0 + a(\alpha_2 - 2a \cdot \alpha_3) + a^2 \cdot \alpha_3 \\ a_1 = \alpha_2 - 2a \cdot \alpha_3 \Rightarrow \alpha_2 = a_1 + 2a \cdot \alpha_3 \\ a_2 = \alpha_3 \Rightarrow \alpha_3 = a_2 \end{cases}$$

$$\text{Maybe } B = (1, x-5, (x-5)^2)$$

$$f = 1+x$$

(3) Det. the number of bases in $\mathbb{Z}_2^3$.

$$\mathbb{Z}_2^3 = \{(\hat{0}, \hat{0}, \hat{0}), (\hat{0}, \hat{0}, \hat{1}), (\hat{0}, \hat{1}, \hat{0}), (\hat{0}, \hat{1}, \hat{1}), (\hat{1}, \hat{0}, \hat{0}), (\hat{1}, \hat{0}, \hat{1}), (\hat{1}, \hat{1}, \hat{0}), (\hat{1}, \hat{1}, \hat{1})\}$$

A basis: $(z_1, z_2, z_3), z_1, z_2, z_3 \in \mathbb{Z}_2^3 \setminus \{\hat{0}, \hat{0}, \hat{0}\}$

↓
it cannot generate itself.

We choose: $z_1 \in \mathbb{Z}_2^3 \setminus \{\hat{0}, \hat{0}, \hat{0}\}$ -

$$\Rightarrow 2^3 - 1 = 7$$

We choose: $z_2 \in \mathbb{Z}_2^3 \setminus \{\hat{0}, \hat{0}, \hat{0}, z_1\} \Rightarrow 2^3 - 1 - 1 = 6$

We choose: $z_3 \in \mathbb{Z}_2^3 \setminus \{\hat{0}, \hat{0}, \hat{0}, z_1, z_2, \langle z_1, z_2 \rangle = (z_1 + z_2)\} \Rightarrow 2^3 - 1 - 2 = 4$

(2^4 does not exist)

Nr. of bases: $\langle z_1, z_2, z_3 \rangle = 7 \cdot 6 \cdot 4 = 168$
(not distinct)

Dimension

Lecture 6

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